

HARDWARE INTEGRATION · CONFIDENTIAL

HearMe

NeuroBand — Bio-Signal & Silent Trigger Safety System

VOLUME IV

Prepared for: Debjyoti Sarkar

Product: HearMe — Women's Personal Safety App

Scope: Wearable BLE band · Bio-signal fusion · Silent SOS

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1. Concept — What NeuroBand Is, and Why Now

The HearMe stack already ships an active-trigger surface area: shake, voice safe-word, duress PIN, timer check-in, in-app SOS button. Every one of them assumes the user can *act*. The **NeuroBand** closes the last gap — the woman who is held still, gagged, or whose phone is just out of reach. Her body still leaks distress: heart rate spikes, skin sweats, skin cools, fingers clench. NeuroBand is a wrist-worn BLE band that captures those signals and hands them to the HearMe phone, which alone decides whether to fire SOS.

This implements **§8 Biometric Wearable Layer** of *HearMe Hardware & IoT Strategy — Volume III*. The strategy doc commits to the “physiological duress signature: any two of the five physiological signals firing together” rule. Volume IV is how it ships.

Architectural rule, unchanged from Volume III: the band is dumb. It samples, filters, and ships features. The phone owns all decision logic. A leaked GATT packet must reveal nothing; a stolen band must trigger nothing.

5

BIO SENSORS

2 of 5

FUSION THRESHOLD

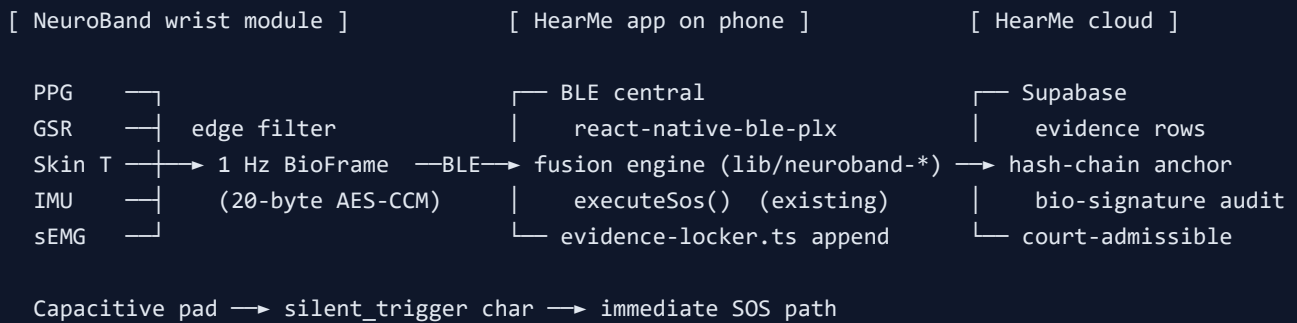
~3 days

BATTERY (1 HZ)

< 800 ms

TRIGGER → SOS

2. System Architecture at a Glance



Every box on the right already exists in the repository today. `executeSos()` in `providers/HearMeProvider.tsx`, the evidence locker, the Supabase sync, the haptic ack pipeline — all reused. The new code is a single BLE bridge plus a fusion module. NeuroBand is additive, not architectural.

3. Hardware Bill of Materials

One band, with rationale. Pricing in INR at qty-100 sourcing (Robu, SiliconCraft, DigiKey IN).

#	Function	Part	Why this part	Qty	₹ / unit
1	MCU + BLE 5.0 radio	Nordic nRF52840 (MDBT50Q-1MV2 module)	Cortex-M4F + crypto coprocessor + USB DFU. Same family as your Tier-1 BLE Clip — identical pairing protocol, shared firmware infra.	1	350
2	PPG — HR & SpO ₂	MAX30102 (or MAX86141 for clinical grade)	Integrated red+IR LEDs and photodiode, I ² C, AGC. Dominant DIY+OEM choice for wrist-worn HR.	1	180
3	GSR / EDA	Grove GSR v1.2 + 2× Ag/AgCl dry pads	Skin conductance is the cleanest panic biomarker. Reads in µS; survives mild perspiration.	1	220
4	Skin temperature	MAX30205 contact (or MLX90615 IR)	±0.1 °C accuracy. Detects shock / freezing / circulatory shutdown.	1	160
5	IMU (motion / fall)	BMI270 or LSM6DSOX	Distinguishes “HR spike while running” from “HR spike while still.” The whole rule depends on this gate.	1	140
6	sEMG (fight-or-flight)	MyoWare 2.0 + 3 dry pads	Wrist-tension arousal — clenched fist sustained > 4 s. Optional in v1; high BOM impact.	1	600
7	Haptic feedback	DRV2605L + 10×3 mm LRA coin	Confirms “SOS sent” and “contact ack” silently. Felt, not seen.	1	90
8	Capacitive override pad	TTP223 + copper pad under strap	Triple-tap silent SOS even if biosignals miss. Backup path when fusion is in workout mode.	1	25
9	Battery	3.7 V 110 mAh LiPo + JST-PH	~3 days at 1 Hz BLE notify, 4 Hz GSR poll. Keeps band thin enough to disappear under a sleeve.	1	90
10	Charging	TP4056 + USB-C breakout	Standard Li-ion charge IC; replace with Qi-Rx coil if you target IP67 with no ports.	1	60
11	Power management	TPS62740 buck + MCP1700 LDO (analog rail)	Separate clean rail for GSR / PPG analog front-end. Shared rail = noise = false positives.	1	70
12	Enclosure	TPU shell + silicone strap, IP67 gasket	Hides electrodes flush with skin. Resin-printed prototype, injection-mold for production.	1	200
13	Status LED	Single RGB indicator	Charge / paired / fault. No loud LEDs — the band must look like a fitness tracker.	1	15

BOM totals. ~₹2,200 / unit at qty-100, ~₹1,400 at qty-5,000. Sells at Tier-3 ₹3,500–₹5,000 in your existing pricing ladder, with healthy D2C margin and headroom for retail markup.

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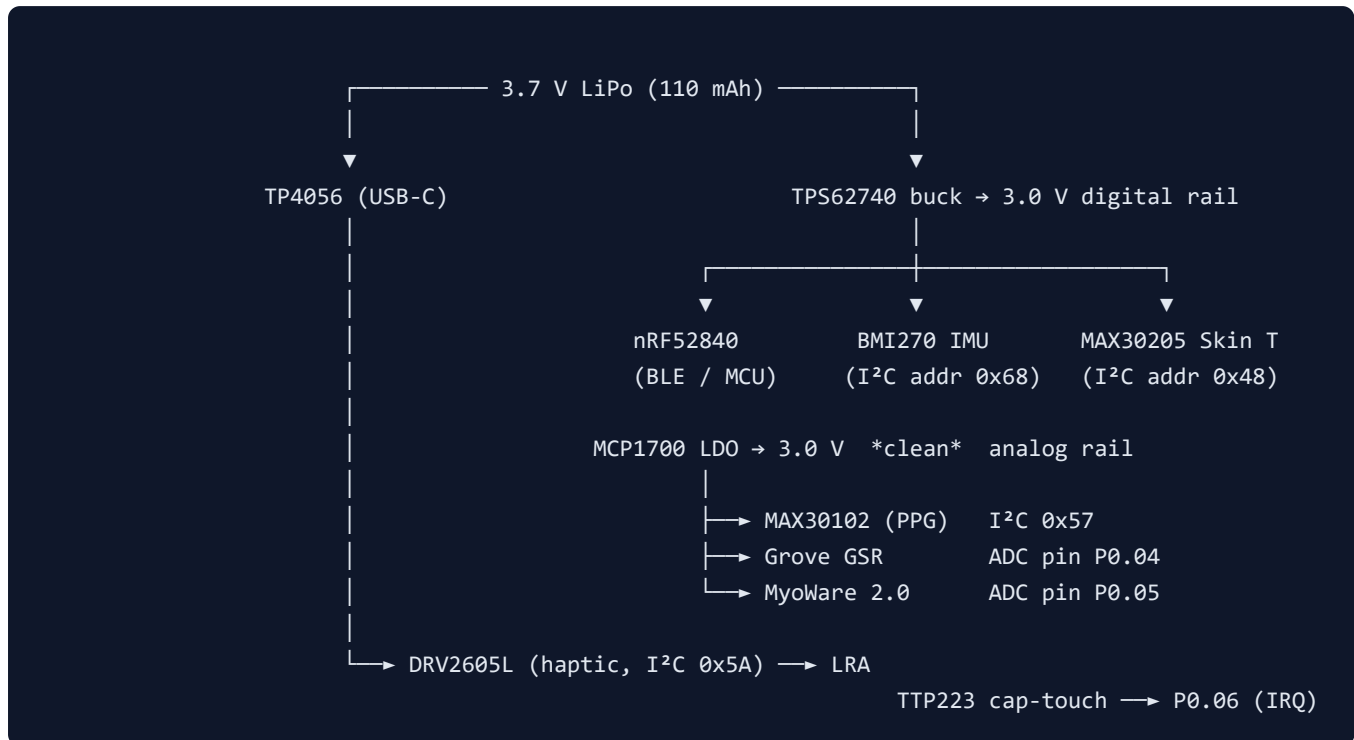
3.1 One-time prototyping tools

Tool	Purpose	Approx ₹
Nordic nRF52840-DK	Primary dev board with onboard J-Link	4,500
J-Link EDU mini (spare)	SWD programmer + RTT logging	2,200
Saleae Logic 8 clone	I ² C / SPI / BLE timing debug	1,800
nRF52840 USB Dongle (Wireshark)	BLE air-trace sniffer	1,800
Polar H10 chest strap	Ground-truth HR for PPG calibration	6,500
BIOPAC GSR / Empatica E4 (loaner)	Ground-truth electrodermal activity	15,000

3.2 Phone-side prerequisite

No new hardware on the phone. BLE 5.0 is universal on every Android since 2018 and every iPhone since 6s. The HearMe app must run on a **custom Expo dev client** — Expo Go cannot load `react-native-ble-plx`. This is a one-time `eas build --profile development`.

4. Reference Schematic & Power Domains



Two separate 3.0 V rails are non-negotiable. The PPG and GSR analog front-ends are easy to poison with switching noise from the BLE radio. Buck for digital, LDO for analog, star-ground at the LiPo – node. Skipping this is the #1 reason DIY bio-bands false-trigger.

5. Firmware Architecture (Zephyr + NimBLE)

Zephyr RTOS with the NimBLE host. Nordic SDK is being deprecated; do not start a new project on it.

```

Sensor task                                10 Hz
├─ MAX30102 → HR, HRV-RMSSD, IBI, SpO2
├─ Grove GSR → tonic + phasic split (μS)
├─ MAX30205 → skin °C
├─ BMI270 → accel, gyro, motion class
└─ MyoWare → sEMG envelope

Edge filter (rolling 30-s window)
├─ HR baseline (Welford online mean+var)
├─ GSR phasic-tonic decomposition
├─ Skin-temp drift vs 5-min average
└─ Motion classifier { still | walk | run }

GATT publisher                            1 Hz notify
└─ packs into the 20-byte HearMe BioFrame

Capacitive override task                  50 Hz poll
└─ writes silent_trigger char on triple-tap

Crypto + counter                          per-frame
└─ AES-CCM, monotonic 32-bit counter

```

The band ships features, not raw waveforms. That keeps BLE bandwidth low, battery healthy, and means a captured GATT packet contains a single byte of HR, not a 250 Hz PPG trace that could fingerprint the wearer.

6. BLE GATT — HearMe Bio Service

Custom 128-bit UUID base, four characteristics. All notify-and-write traffic is AES-CCM encrypted using a per-session key derived during pairing.

Characteristic	UUID suffix	Direction	Payload
<code>bio_frame</code>	<code>...8001</code>	Notify, 1 Hz	20 B: <code>[seq u16][hr u8][hrv_rmssd u8][spo2 u8][gsr_us u16][temp_c10 i16][motion u8][semg u8][battery u8][flags u8][counter u32]</code>
<code>silent_trigger</code>	<code>...8002</code>	Notify	1 B: <code>bit0=triple-tap, bit1=auto-fired, bit2=tamper</code>
<code>command</code>	<code>...8003</code>	Write	Calibrate / haptic-buzz / OTA-handoff / sleep
<code>device_info</code>	<code>...8004</code>	Read	Firmware version + serial (no PII)

Field encoding rationale: HR fits in `u8` up to 255 bpm; skin temp is stored as $^{\circ}\text{C} \times 10$ in `i16` for $\pm 0.1^{\circ}\text{C}$ resolution, GSR in $\mu\text{S} \times 100$ as `u16` covers 0–655 μS which spans every realistic skin response. The trailing `u32 counter` is the monotonic value Volume III mandates for replay protection on every GATT write.

7. Pairing, Crypto & Replay Protection

1. Each band ships with a unique 12-character **PSK printed on a sticker** on the inside of the strap.
2. HearMe app scans, finds the band, prompts for the PSK.
3. App and band perform **ECDH key exchange** over BLE, mixing the PSK as a salt to defeat MITM.
4. The derived 128-bit session key feeds **AES-CCM** on every frame thereafter.
5. The 32-bit counter increments per frame; the phone rejects any frame whose counter is $\leq$ the last accepted value.
6. Re-pair clears the counter and rotates keys.

No “0000”-class PINs. Volume III explicitly forbids weak pairing. PSK + ECDH is the only acceptable scheme; reuse the implementation you ship for the Tier-1 BLE Clip so a single audit covers both products.

8. Signal Fusion — The 2-of-5 Duress Rule

This is the only piece of intelligence on the phone. It must live in the app, not in firmware, so you can A/B-test thresholds across the user base without re-flashing every band.

```
markers per 30-s window (each is 0 or 1):

M_hr      = HR > baseline + 25 bpm   AND   motion ∈ { still, walk }
M_gsr     = GSR phasic peak > 1.5 × tonic-baseline
M_temp    = skin-temp drop > 0.4 °C from 5-minute average
M_spo2    = SpO2 < 92 %                (crash / unconsciousness fallback)
M_semg    = sEMG envelope > 2 σ above resting (clenched fist sustained > 4 s)

if  popcount(markers) >= 2
  AND not in_workout_mode
  AND time_since_calibration > 24 h:
    fire silent SOS
```

Marker	Sensor	Why it fires during distress
M_hr	PPG	Sympathetic surge — fear response spikes HR even when the person is held still.
M_gsr	GSR	Sweat-gland activation is the most reliable panic biomarker; can't be voluntarily suppressed.
M_temp	Thermopile	Peripheral vasoconstriction during shock drops skin temp before core temp moves.
M_spo2	PPG (red+IR)	Falls when restraint compromises breathing or after physical assault.
M_semg	sEMG	Clenched-fist tension — the involuntary fight-or-flight muscle response.

The motion gate is what makes this safe to ship. Without it, two markers will fire constantly during a run. The Volume III rule explicitly says “heart-rate spike *with no motion*.” Honour it.

9. Phone-Side Integration into the HearMe App

Your repo already has every adjacent piece. The integration is one BLE bridge plus one fusion module wired into the existing `executeSos()` path.

9.1 Dependencies (custom dev client only)

```
npx expo install react-native-ble-plx
npx expo install expo-dev-client
npx expo prebuild
eas build --profile development      # one-time per platform
```

Add to `app.json`:

```
"plugins": [
  ["react-native-ble-plx", {
    "isBackgroundEnabled": true,
    "modes": ["central"],
    "bluetoothAlwaysPermission":
      "HearMe pairs with your NeuroBand to detect distress."
  }]
]
```

9.2 New files (matching your existing `lib/*` style)

Path	Responsibility
<code>lib/neuroband-ble.ts</code>	Connect, subscribe, decode BioFrame, expose a typed stream.
<code>lib/neuroband-fusion.ts</code>	Compute markers, run the 2-of-5 rule, emit a <code>'duress' 'idle'</code> verdict.
<code>lib/neuroband-pairing.ts</code>	PSK + ECDH using <code>expo-crypto</code> ; persists session key in <code>expo-secure-store</code> .
<code>lib/neuroband-storage.ts</code>	Serial number, baselines, calibration state, last-seen timestamp.

9.3 Provider wiring

Mirror the existing `ShakeBridge` pattern in `providers/HearMeProvider.tsx`:

```
function NeuroBandBridge({ enabled, onDuress, onSilentTrigger }: Props) {
  useEffect(() => {
    if (!enabled) return;
    const sub = NeuroBand.subscribe((frame) => {
      const verdict = fuse(frame); // 2-of-5 rule
      if (verdict === 'duress') onDuress();
      if (frame.flags.tripleTap) onSilentTrigger();
    });
    return () => sub.remove();
  }, [enabled, onDuress, onSilentTrigger]);
  return null;
}
```

Both handlers route into the existing `executeSos()` — but with `skipSosConfirm = true` and `sirenEnabled = false` for genuinely silent operation. **Do not open an Alert dialog** on duress — that defeats the entire point.

9.4 Settings additions (`lib/types.ts`)

```
export type HearMeSettings = {
  // ...existing fields
  neuroBandEnabled: boolean;
  neuroBandSerial: string | null;
  neuroBandWorkoutModeUntil: number | null;
  neuroBandCalibratedAt: number | null;
  neuroBandSensitivity: 'low' | 'normal' | 'high';
};
```

PSK and session key live in `expo-secure-store` (same pattern as `lib/duress.ts` uses for the duress PIN). Never put them in `HearMeSettings`.

9.5 New screen

`app/(main)/neuroband.tsx` — pair / unpair, live signal dashboard, calibration progress, workout-mode toggle. Match the visual language of `behavior-monitor.tsx` and `journey-monitor.tsx`. Surface it from `app/(main)/(tabs)/safety.tsx` as a card next to Voice Trigger.

9.6 Background operation

- **Android:** declare `FOREGROUND_SERVICE_CONNECTED_DEVICE` and show a persistent “NeuroBand monitoring” notification.
- **iOS:** add `bluetooth-central` to `UIBackgroundModes`; iOS will keep the app suspended-but-resumable on BLE notify.

10. Evidence Trail & Cloud Sync

When NeuroBand fires SOS, append a `'bio'` evidence item alongside the existing location stamp. Extend `lib/evidence-locker.ts` :

```
session = addEvidenceItem(session, {
  type: 'bio',
  uri: null,
  text: `NeuroBand duress: HR=${hr} bpm, GSR=${gsr} μS, ` +
    `temp=${temp.toFixed(1)} °C, motion=${motion}`,
  lat, lon,
  alertId: null,
  tags: ['sos', 'neuroband', `markers:${marker_bitmap}`],
});
```

This gives you the legal chain — physiological signature → timestamp → location → cloud sync → hash-chain anchor — already documented as court-admissible in Volume III §11. The marker bitmap (e.g. `markers:01101`) is what forensics will care about: it explains *why* the system fired, not just that it fired.

11. Calibration, Workout Mode & False-Positive Control

Mechanism	What it does	How it's exposed
24-hour calibration window	For the first 24 h after pairing the band streams data but cannot fire SOS. Establishes per-user HR / GSR / temp baselines.	"Learning your body" onboarding card on the NeuroBand screen.
Workout mode	Disables the fusion engine for a user-set duration (default 1 h). Cap-touch override still works.	One-tap toggle; auto-prompts when IMU detects sustained run pattern for > 5 min.
Sensitivity dial	Switches threshold from 2-of-5 (normal) to 3-of-5 (low) or HR±15 bpm (high).	<code>neuroBandSensitivity</code> in Settings.
Confirm-window for borderline events	If exactly 2 markers fire but motion class is ambiguous, the band buzzes once and waits 10 s for the user to swipe-cancel.	Haptic; auto-fires if no input. Volume III calls this "reassurance loop — felt, not seen."
Adverse-event log	Every fire / cancel / mis-fire records a row to a local debug log. Used to retune thresholds.	Hidden behind a long-press on the NeuroBand screen header.

12. Privacy, Security & Regulatory Posture

- PSK + ECDH pairing, AES-CCM on every frame, monotonic 32-bit counter — *not* a fixed PIN.
 - The band stores **no PII**. Serial number only. A confiscated band reveals nothing about its wearer.
 - Firmware OTA images are signed with the HearMe `ed25519` key — same key already mandated for the BLE Chip.
- BLE Chip

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- Raw waveforms **never leave the band**. Only computed features. A subpoena to the cloud cannot return PPG traces because they were never uploaded.
- The HearMe app shows a persistent “NeuroBand is monitoring” status whenever the engine is armed. Hidden surveillance is a liability, not a feature.
- Bio-data classification: under India’s DPDP Act 2023 this is sensitive personal data. Store in Supabase with row-level security keyed to the user’s auth ID, encrypted at rest.
- Medical-device positioning: the band is a **safety wearable, not a medical device**. The app must not advertise diagnostic claims (e.g. “detects atrial fibrillation”) without CDSCO clearance.

13. Build, Test & Validation Order

1. **Wire the breadboard.** nRF52840-DK + MAX30102 + Grove GSR + BMI270 on jumpers. Skip the enclosure entirely until step 7.
2. **Stream BioFrame to nRF Connect mobile.** Verify the 20-byte struct decodes cleanly. Walk around, check counter increments, check no dropped notifies.
3. **Add `lib/neuroband-ble.ts`** with a debug screen that just plots live HR / GSR. Ignore fusion; this is plumbing only.
4. **Wear the breadboard for 24 h yourself.** Capture HR baselines (rest, walk, run) and GSR baselines (cool, warm, sweaty). This is the dataset that tunes the thresholds.
5. **Tune the fusion thresholds** with the captured data. The values printed in §8 are starting points, not gospel. Expect to move them $\pm 20\%$.
6. **Stage the duress trigger** with a benign side-effect first: haptic buzz + console log. *Do not wire `executeSos()` yet.*
7. **Wire the real SOS path.** Run the pre-flight: workout-mode lockout, calibration check, motion gate, sensitivity dial. Test with a tame contact list.
8. **Add the capacitive override** last. It's the easiest path; if it's present in step 1 you'll lean on it and never validate fusion.
9. **PCB spin** only after fusion accuracy is stable on breadboard. EasyEDA + JLCPCB ~₹6,000 for 5 prototypes.
10. **Field pilot** with 10 users for 2 weeks. Collect adverse-event log. Re-tune. Then ship.

14. Cost Sheet & Order List

14.1 Per-band BOM (qty 100)

Category	Subtotal ₹
MCU + radio (nRF52840 module)	350
Bio-sensor stack (PPG + GSR + temp + IMU + sEMG)	1,300
Haptic + cap-touch	115
Power (battery + charger + buck + LDO)	220
Enclosure + strap + LED	215
Total per band	~ 2,200

14.2 Smallest viable starter cart

One-time outlay to get a working prototype on the bench this week:

Item	₹
nRF52840-DK	4,500
J-Link EDU mini (spare)	2,200
MAX30102 breakout	180
Grove GSR v1.2 + Ag/AgCl pad pack (50)	220
MAX30205 breakout	160
BMI270 breakout	140
DRV2605L + 10×3 mm LRA	90
2× 3.7 V 110 mAh LiPo	180
TP4056 USB-C charger module	60
Jumper wires, 0.1" headers, breadboard	300
Starter total	~ 8,030

Vendors: **Robu.in**, **SiliconCraft**, **QuartzComponents** for India stock; **DigiKey IN** for the nRF52840 module if local supply is dry.

15. 12-Week Delivery Roadmap

Week	Deliverable	Outcome
1–2	Breadboard build, BioFrame streaming to nRF Connect	End-to-end BLE link proven
3	<code>lib/neuroband-ble.ts</code> + debug plot screen in HearMe	App reads live HR / GSR; 24 h self-wear baselines captured
4	Fusion module + threshold tuning	2-of-5 rule firing on demand against staged scenarios
5	PSK + ECDH pairing flow + secure-store wiring	Production-grade crypto, parity with Tier-1 BLE Clip
6	Provider integration + new screen + Settings additions	Feature-complete on the phone side
7	Capacitive override + workout mode + sensitivity dial	False-positive controls all in place
8	Evidence-locker <code>'bio'</code> type + cloud sync	Court-admissible chain extends to bio-signature
9	PCB v1 layout, JLCPCB order, 5 prototype boards	Form factor moves from breadboard to wrist
10	Enclosure print + battery integration + IP67 gasket	Wearable prototype in hand
11	10-user closed pilot, adverse-event collection	Real-world false-positive rate measured
12	Threshold re-tune + firmware OTA pipeline + soft launch	NeuroBand v1 ready for D2C / NGO partnership ship

What to do today

Place the ₹8,030 starter-cart order. While it's in transit, scaffold `lib/neuroband-ble.ts` and the new screen so the moment the parts land you can flash, pair, and see live HR on the phone the same evening. NeuroBand is the wearable that finally lets HearMe protect the user when she cannot reach for her phone — ship it.